INCH-POUND

MIL-T-24944(NU)
7 June 1989

MILITARY SPECIFICATION

TROUSERS, FLIGHT DECK, CREWMEN'S

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for flight deck trousers made from a Flame Retardant Treated 100% cotton denim material.

1.2 Classification. The trousers shall be in the following sizes and lengths (see 6.2).

1.2.1 Sizes. 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 40, 42, 44, 46, 48, and 50.

1.2.2 Lengths. Regular and X-Long.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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SPECIFICATIONS

FEDERAL

V-F-106	- Fasteners, Slide, Interlocking
NN-P-71	- Pallets, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way (Partial)
DDD-L-20	- Label; For Clothing, Equipage and Tentage, (General Use)

MILITARY

MIL-P-15011	- Pallet, Material Handling, Wood Post Construction, 4-Way Entry
MIL-B-17757	- Box, Shipping, Fiberboard (Modular Sizes)
MIL-C-24915	- Cloth, Denim, Cotton (Flame Retardant Treated)
MIL-F-10884	- Fastners, Snap
MIL-F-21840	- Fastener Tapes, Hook and Loop, Synthetic
MIL-T-43548	- Thread, Polyester Core: Cotton-, Rayon- or Polyester-Covered

STANDARDS

FEDERAL

FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-1488	- Provisions for Evaluating Quality of Trousers
MIL-STD-2073-1	- DoD Materials Procedures Development and Application of Packaging Requirements
MIL-STD-2073-2	- Packaging Requirement Codes

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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LAWS AND REGULATIONS

U.S. Postal Service Manual

(Copies of the manual should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036-1404.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Applications for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606-5808.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified, (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.3 Material.

3.3.1 Basic material. The basic material for the trousers shall be a Flame Retardant Treated cotton denim conforming to MIL-C-24915. The material shall be dyed shade Blue 3371.

3.3.2 Thread.

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3.3.2.1 Thread, polyester, core. The thread for the trousers shall be a cotton-covered or rayon-covered polyester core thread, two or three ply, ticket No. 50 or 70 of MIL-T-43548, except that the polyester-covered polyester core thread shall not be used. The thread shall be dyed to approximate the shade of the basic material.

3.3.2.2 Colorfastness. The dyed thread shall show fastness to light and laundering equal to or better than the standard sample when tested as specified in MIL-T-43548. As a limit of acceptability or when no standard sample is available, the dyed thread shall show a minimum of "good" fastness to light and laundering when tested as specified in MIL-T-43548.

3.3.3 Slide fastener. The slide fastener for closing the fly shall conform to type I, style I, cam lock or semi automatic pin lock, size LMS, short tab pull of V-F-106. All metal components of the slide fastener shall be fabricated of brass except for the semi automatic pin lock cover and spring, which shall be stainless steel. All components shall be from the same supplier. The slide fastener shall be furnished with a staple bottom stop (see 4.4.1).

3.3.3.1 Tape for slide fastener. The tape for the trouser slide fastener shall be high temperature resistant aramid. The color of the tape shall be black, and shall meet commercially acceptable colorfastness requirements for aramid tapes.

3.3.4 Snap fastener. The snap fasteners for the waistband shall be brass, conforming to style 2A, finish 2, male and female complete, consisting of stud with size 1 or 2 eyelet and socket with size 1 or 2 eyelet (instead of button) conforming to MIL-F-10884.

3.3.5 Fastener Tapes. The fastener tapes shall be Flame Retardant Treated conforming to type II, class 1 of MIL-F-21840. The color shall be Black 3230. Unless otherwise specified, the tapes shall be 1 inch in width.

3.3.6 Labels.

3.3.6.1 Combination label. Each pair of trousers shall have a combination identification, personal and instruction label conforming to type VI, classes 10 and 15 of DDD-L-20, except that the label shall be folded. When attached, (see operation 16b) the identification portion of the label shall be on the side of the label facing the wearer and the laundering instructions shall be on the reverse side. The printing shall show fastness to laundering and shall bear the following inscription and format:

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NAME:
SERVICE NO:
TROUSERS, FLIGHT DECK, CREWMEN'S
FIBER CONTENT: 100% COTTON (FLAME RETARDANT TREATED)
CONTRACT NO. DLA-100-00-0-0000 (EXAMPLE)
NAME OF CONTRACTOR:
NAME OF MANUFACTURER: (IF OTHER THAN CONTRACTOR)

LAUNDERING INSTRUCTIONS

FILL WASHER AND DRYER TO 2/3 CAPACITY
GARMENTS MUST ALWAYS BE TURNED INSIDE OUT
MACHINE WASH, HOT WATER,
NO STARCH, NO BLEACH
TUMBLE DRY, MEDIUM HEAT
REMOVE PROMPTLY

OR

FOLLOW NAVEDTRA 414-01-45-81, FORMULA II
FILL WASHER AND DRYER TO 2/3 CAPACITY
GARMENTS MUST ALWAYS BE TURNED INSIDE OUT

The printed label should use uppercase letters, bold print and underlining as indicated above.

3.3.6.2 Size label. The size label for the trousers shall conform to type IV or VI, class 2 of DDD-L-20 and shall bear the following inscription:

Size 38R (Example)
STOCK NO: 8415-00-000-0000 (Example)

NOTE: The lengths (Regular and X-Long) may be abbreviated as R or XL

3.4 Design. The trousers shall be of double needle construction, fabricated from denim fabric. The trousers shall feature two cargo pockets, two knee patches, two back waist darts, snap fastener waistband, slide fastener fly closure, and separate one-piece waistband with belt loops. The leg style shall be straight and unhemmed (see Figure 1).

3.4.1 Figure. The figure is furnished for information purposes only. When inconsistencies exist between the written specification and the figure, the written specification shall govern.

3.5 Patterns. Standard patterns for the trousers will be furnished by the Government to the contractor for use in cutting the working patterns (see 6.3). The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way, except that additional notches for use during construction are permitted on the working patterns. Also, minor modifications are permitted where necessary when using automatic equipment. These modifications shall not alter the dimensional, serviceability or appearance requirements cited in the specification. The patterns allow 1/2 inch seam allowance for all seams, except as otherwise indicated in Table I.

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3.5.1 Pattern parts. The component parts of the trousers shall be cut from material as specified and in accordance with the pattern parts required for the manufacturing process as follows:

Material	Nomenclature of pattern parts	Cut parts
Basic material	Backs	2
	Fronts	2
	Waistband	1 1/
	Right fly	1
	Left fly facing	1
	Right fly lining	1 2/
	Cargo pockets	2
	Cargo pocket flaps	2
	Knee patch	2
	Leg tab	2

NOTE: Pattern is not furnished for belt loops (see operation 7).

1/ Waistband may be banded (see operation 16).
2/ May be cut either face up or face down.

3.6 Construction.

3.6.1 Stitches, seams, and stitching. Stitch, seam, and stitching types specified in Table I shall conform to FED-STD-751. Whenever two or more methods for seam or stitch types are given for the same operation any one may be used. Seam allowances shall be maintained with seams sewn so that no raw edges, runoffs, twists, pleats, puckers or open seams result. All seams shall start and finish evenly. Thread tension shall be maintained so that there is no tight or loose stitching. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the trousers. Ends of all seams produced by stitch type 401 shall be caught in other seams or stitchings. The width of the safety stitch shall be 3/8 inch and the guides and knives on the safety stitch machine shall be set to trim only the ravelled ends of the fabric. Unless otherwise specified, topstitching and edgestitching shall be 1/16 to 1/8 inch from the edge and all double needle stitching shall be 1/4 inch gauge with the row of stitching next to the folded edge not more than 1/16 inch from the edge. The use of a trimming knife in the overedging of the right fly in operations 9f and 9g is prohibited.

3.6.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table I.

3.6.3 Thread breaks and ends of seams. Ends of all seams and stitchings when not caught in other seams or stitchings shall be backtacked not less than 1/4 inch. The ends of a continuous line of stitching (except label) shall be overlapped not less than 1/2 inch. The ends of the label stitching shall be overlapped not less than 3 stitches. Thread breaks (all stitch types) shall be secured by stitching back of break not less than 1/2 inch. Skipped stitches or thread breaks (401 stitch type) may be repaired by using 301 stitch type. A seat seam shall be completely resewn. The ends of type 401 stitching not caught in other seams or stitching shall have a 1/2 to 3/4 inch chain

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extending from it.

3.7 Manufacturing operations requirements. The trousers shall be manufactured in accordance with operation requirements specified in Table I. The contractor is not required to follow the exact sequence of operations listed provided that the finished trouser is identical to that produced by following the sequence of operations listed in Table I. Minor modifications are permitted where necessary when using automatic equipment. These modifications shall not alter the dimensional, serviceability, or appearance requirements cited in the specification.

3.7.1 Shade and size marking. The component parts of the trousers shall be marked, ticketed, or bundled to insure a uniform shade and size throughout the garment. Any method may be used except:

(1) Corrosive metal fastening devices.
(2) Sew-on shade tickets.
(3) Adhesive type tickets which discolor or adhere to the material upon removal of the tickets.

NOTE: The use of ink pad numbering machine, rubber stamp or pencil is allowed, provided the numbering does not show on the outside of the garment and wherever possible, is covered by the seam allowance.

3.7.2 Pressing. All pressing required in Table I shall be done with heated pressing iron or machine.

3.7.3 Abbreviations in table of operations. The abbreviations used in Table I are as follows:

Stch	-	Stitch
in	-	inch
Ndl	-	Needle
Bob	-	Bobbin
Lpr	-	Looper
Mchne	-	Machine
Brtck	-	Bartack
smlr	-	similar
incl	-	including
dbl	-	double
tckr	-	tacker
bx stch	-	box stitch

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TABLE I - CONSTRUCTION OF TROUSERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
1.	Cutting. a. Cut the trousers in strict accordance with patterns furnished, which shall show size, placement of component parts, directional lines for cutting, and notches for the proper assembly of all component parts. b. All component parts shall be cut from one piece of material, except the right fly, left fly facing, right fly lining, banded waistband, leg tabs and belt loops, which may be cut from ends. Parts cut from ends shall approximate the shade of the trousers. c. Lay fronts and backs so that the directional line is in the warp direction. The directional line may vary from the warp direction by not more than 1-1/2 inches on both fronts and backs. Measurement shall be taken from the top and bottom of the directional line on pattern to the selvage edge of the fabric and the difference between the two measurements shall not exceed the tolerance specified. d. Cut the stripping for belt loops and banded waistband of sufficient width to comply with operations 7 and 16. The stripping may be placed on rolls, however, there shall not be any pieced belt loops or waistband visible on the finished trousers.					
2.	Replacement of defective components. During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in MIL-STD-1488 shall be removed from production and replaced with nondefective and properly matched components.					
3.	Shade and size marking (see 3.7.1). All parts shall be marked, ticketed or bundled, except those parts cut from ends.					

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TABLE I - CONSTRUCTION OF TROUSERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
4.	Make darts.					
	a. Fold darts as indicated by marks on patterns and stitch tapering lower end of dart to nothing. Thread ends at lower end of darts shall not be trimmed closer than 1/2 inch in length or when automatic dart machine is used, the thread ends at lower end of dart shall be backstitched.	301 or 401	OSf-1	8-10	50 50	50 70
	b. Fold darts inside towards center back seat seam and edge stitch to back parts, 1/16 inch from folded edge. NOTE: When drill marks are used, they should be placed 1/2 inch above bottom of dart, and be covered by the stitching.	301	OSe-2	8-10	50	50
5.	Assemble and attach leg tabs and take-up tape.					
	a. Overedge one long side of leg tab.	503 or 504 or 602	EFd-1	6-10	70	70
	b. Fold edges of leg tab in thirds in accordance with pattern marks with the overedged side visible. Stitch around all sides through all plies, 3/16 to 1/4 inch from folded edges.	301	LSbj-1	8-10	50	50
	c. Fold under one end of leg tab 1 to 1 1/8 inch. Position a 2 to 2 1/8 inch length of 1 inch hook fastener tape to underside of the leg tab, according to pattern marks, covering folded end. Stitch to leg tab, 1/16 to 1/8 inch from edges.	301	EFa and LSbj-1	8-10	50	50
	d. Apply a 1 1/4 to 1 3/8 inch length of 1 inch loop fastener tape to the front of each trouser leg, as indicated by marks on pattern and stitch around all sides, 1/16 to 1/8 inch from edges.	301	LSbj-1	8-10	50	50
	e. Align hook portion of leg tab onto loop portion so that the leg tab is parallel to the trouser bottom and the free end faces toward the inseam. Fold under the free end of leg tab 1/4 to 5/16 inch and stitch to trouser leg with a box-stitch (approximately 1 X 3/4 inch) through all plies, 3/16 to 1/4 inch from edges.	301	bx stch	8-10	50	50

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No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
	f. The finished dimensions of the leg tabs shall be 5 1/4 to 5 3/8 inches X 1 1/8 to 1 3/16 inches.					
	g. Position a 3 to 3 1/8 inch length of 1 inch loop fastener tape on to the back of each leg in accordance with pattern marks. Stitch around all sides, 1/16 to 1/8 inch from edges.	301	LSbj-1	8-10	50	50
6.	Make cargo pocket flaps.					
	a. Stitch two 1 1/4 to 1 3/8 inch lengths of 1 inch hook fastener tape to each pocket flap as indicated by pattern marks. Stitch 1/16 to 1/8 inch from edges.	301	LSbj-1	8-10	50	50
	b. Fold flap in half, face to face and stitch left and right side seams, 1/4 (± 1/16) inch from edges.	301	SSe-2 (a)	8-10	50	50
	c. Turn flap, work out points and topstitch 1/4 inch from the folded edges along the sides and bottom.	301	SSe-2 (b)	8-10	50	50
7.	Make belt loops.					
	a. Fold stripping with edges abutted at center and double stitch not less than 3/16 inch gauge, using a cover stitch on the underside.	406	EFh-1	6-10	50	50
	or					
	b. Fold stripping lengthwise. Turn under both raw edges and stitch each edge 1/16 to 1/8 inch from the edge.	301 or 401	EFp-2	8-10	50 50	50 70
	c. The finished loops shall measure 7/16 to 1/2 inch wide.					
8.	Make cargo pockets.					
	a. Overedge stitch top edge of cargo pocket.	503 or 504 or 602	EFd-1	6-10	70	70

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No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
	b. Fold top of pocket inward according to pattern marks and stitch 1/4 (± 1/16) inch from edge.	301	EFa-1	8-10	50	50
	c. Position two 1 1/4 to 1 3/8 inch lengths of 1 inch loop fastener tape to each pocket as indicated by pattern marks. Stitch 1/16 to 1/8 inch from edges.	301	LSbj-1	8-10	50	50
	d. Fold pocket according to marks on pattern and stitch each of the 2 folds, 1/16 to 1/8 inch from folded edge, forming bellows.	301	OSf-1	8-10	50	50
9.	Make flies.					
	a. Overedge back edge of left fly facing. Operation may be done during 16h.	503 or 504 or 602	EFd-1	6-10	70	70
	b. Double stitch fastener tape 3/16 to 1/4 inch gauge to left fly facing. The edge of the chain shall be 5/8 (± 1/8) inch from front edge of finished fly from top to bottom.	301 or 401	SSau-2	8-10	50 50	50 70
	c. Stitch the right fly to the right fly lining, face to face, or face of right fly to back of right fly lining as applicable, along outward curved edge, 1/4 inch from edge. Turn to finished position and topstitch 1/16 to 1/8 inch from edge.	301	SSe-2	8-10	50	50
	d. Overedge stitch front edge and bottom of right fly lining extension piece.	503 or 504 or 602	EFd-1	6-10	70	70
	e. Stitch fastener tape to back edge of right fly through all plies.	301 or 401	SSa-1	8-10	50 50	50 70
	f. Overedge stitch back edge of right fly catching fastener tape in stitching. This operation is to be performed without the use of a trimming knife.	503 or 504 or 602	EFd-1	6-10	70	70
	or					

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No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
	g. As an alternate, operations 9e. and f. may be performed in one operation by simultaneously stitching and overedging. This operation is to be performed without the use of a trimming knife.	515 or 516 or 519	SSa-1	6-10	50 70	70 70
10.	Attach flies to fronts.					
	a. Stitch left fly facing to left front 1/4 inch from raw edge, face to face according to notches. Turn seam allowance against left fly facing and topstitch 1/16 to 1/8 inch along folded edge of fly.	301	LSq-2	8-10	50	50
	b. Double stitch right fly to front edge of right front from fly notch to top of front.	301 or 401	LSb-2	8-10	50 50	50 70
	c. Attach the slider, and with the scoops evenly matched, staple the bottom of the slide fastener tape. The staple shall be perpendicular to the line of the scoops and shall overlap the scoops. The stop shall be within 3/8 inch of fly notch. The end of the tape shall extend at least 1/8 inch below notch.					
11.	Make and attach knee patches.					
	a. Fold top and bottom edges of knee patches inward 1/2 (± 1/16) inch. Double stitch top and bottom to trousers, 1/4 inch gauge, according to pattern marks.	301	SSb-2	8-10	50	50
12.	Join crotch and seat seam.					
	a. Double stitch crotch seams with left front lapped over right front, with edges folded in, catching right fly extension pieces in the stitching, simulating a double lapped, double stitched seam.	301 or 401	EFa-1 and smlr to LSav-2	8-10	50 50	50 70
	or					

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No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
	b. Turn crotch seam allowance of right fronts to the outside and stitch. Turn edge of left front seam allowance to inside and double stitch, catching the right front crotch seam and right fly extension piece with both rows of stitching. The simulated double lapped seam shall have the left front lapped over right front.	301 or 401	EFa-1 and smlr to LSav-3	8-10	50 50	50 70
	c. Join seat seam with a double lapped and double stitched seam with right back lapped over left back.	301 or 401	LSc-2	8-10	50 50	50 70
13.	Join outseams.					
	a. Stitch outseams with a double lapped and double stitched seam with fronts lapped over back catching ends of knee patches in the stitching.	301 or 401	LSc-2	8-10	50 50	50 70
14.	Attach cargo pocket and flaps.					
	a. Fold under sides and bottom edges of cargo pockets 3/8 to 7/16 inch. Center over left and right side seams according to marks on patterns. Stitch to trousers, catching bottom corner of fold (see operation 8d.), 1/8 inch from edges.	301 or 401	LSc-1	8-10	50 50	50 70
	NOTE: Care shall be taken so that only the bottom corner of the fold is caught in the stitching and that the vertical folds are fully extended when the pocket is opened. The bellows shall face toward the back of the trousers.					
	b. The finished cargo pockets shall measure 7 1/8 (± 1/4) inches in length by 8 1/8 (± 1/4) inches in height.					
	c. Position pocket flaps to trouser fronts according to marks on patterns and stitch 3/16 to 1/4 inch from the raw edge.	301	LSbl-2(a)	8-10	50	50

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	d. Turn pocket flaps down and topstitch 5/16 inch from the folded edge.	301	LSbl-2(b)	8-10	50	50
	e. The finished flaps shall measure 7 3/8 (± 1/4) inches in length by 2 1/2 (± 1/4) inches in height.					
15.	Join inseams. Stitch inseams with a double lapped and double stitched seam with fronts lapped over back catching ends of knee patches in the stitching. The seat seam and crotch seam shall not be out of alignment by more than 3/8 inch.	301 or 401	LSc-2	8-10	50 50	50 70
16.	Attach waistband and labels. a. The top of the slide fastener tape shall be even with the top of unfinished trousers.					
	NOTE: When chain teeth are removed, they shall be within 1/8 inch from lower topstitching of waistband. When they finish greater than 1/8 inch, a top stop shall be required, located not more than 1/16 inch from top of teeth. The top stop shall be located not more than 1/4 inch from lower topstitching of waistband.					
	b. Position waistband on inside top of trousers, aligning raw edges. Stitch 3/8 to 1/2 inch from raw edges simultaneously catching the top edge of combination label (see 3.3.6.1) positioned 1 1/4 (± 1/8) to the left of right outseam and the top edge of size label, centered over the inside of left dart, in the stitching.	301 or 401	BSg-2(a)	8-10	50 50	50 70
	NOTE: The combination label shall be folded in half so that the identification portion of the label shall face the wearer and the instruction portion shall face inward. The stitching shall not show through the printing.					
	c. Turn waistband to finished position, turn bottom edge under 3/8 to 1/2 inch and stitch to top of trousers.	301	BSg-2(b)	8-10	50	50

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	d. As an alternate, waistband can be banded by using a folder and double needle machine.	301	BSc-2	8-10	50	50
	NOTE: The stitching along the bottom edge of the waistband shall be double needle stitching, 1/4 inch gauge. The bottom row of stitching shall be 1/8 inch from the edge. Stitch along the top edge of the waistband with a single row of stitching, 1/8 inch from the edge.					
	e. As an alternate, the size label may be legibly stamped on the left inside waistband, centered over the left dart. The printing shall be legible without smears.					
	f. Center and attach a male snap fastener to right side of waistband so that it is centered (± 1/8 inch) over the right slide fastener teeth.					
	g. Attach a female snap fastener to left side of waistband to correspond with male snap fastener, ensuring a smooth closure when engaged. The socket and stud shall finish on the outside of the waistband.					
	NOTE: When the socket and stud are engaged on the finished trousers, the closure shall not bulge, pucker, twist or be distorted. No metal shall be visible on the outside of the trouser nor shall any metal be against the wearer.					
	h. Overedge stitch left and right ends of waistband (see 9a).	503 or 504 or 602	EFd-1	6-10	70	70

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TABLE I - CONSTRUCTION OF TROUSERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
	i. Fold front end of left waistband and fly to inside and stitch across top edge of waistband extension and down inner edge of fly, from top of waistband to fly notch, forming "j" stitching.	301	LSa-1	8-10	50	50
	j. Fold front end of right waistband to inside and stitch across top edge of waistband extension and down width of waistband along edge, 1/8 inch from the edge.	301	LSa-1	8-10	50	50
	k. The stitching above the curve shall be 1-1/2 (+ 1/4, - 1/8) inches from front edge for all sizes.					
	l. The finished width of the waistband shall be 1-1/2 (± 1/8) inches.					
17.	Attach beltloops. a. The trousers shall have seven belt loops positioned as follows: - One loop centered (± 1/2) inch on each front part - One loop on each side seam - One loop centered (± 1/2) inch on each back part - One loop centered on seat seam					
	b. Place belt loops in desired position (see a. above), underside out with raw edge even with bottom of waistband and bartack 3/16 to 5/16 inch from raw edge.	Brtck or auto loop tckr		21-28 per Brtck 24-28 per Brtck	 50	 50
	c. Turn to finished position, turn under top of belt loops, even with top of waistband and bartack 1/16 to 1/8 inch through all plies of waistband. or	Brtck or auto loop tckr		21-28 per Brtck 24-28 per Brtck		

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TABLE I - CONSTRUCTION OF TROUSERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
	d. As an alternate, turn under top of belt loop 3/8 to 7/16 inch with the folded edge of loop even with the top edge of waistband and bartack through all plies. The bottom free end shall be turned under 3/8 to 7/16 inch and bartacked through all plies. Bartack shall not be more than 1/8 inch from the top and bottom folded ends of loops.	Brtck or		21-28 per Brtck	50	50
		auto loop tckr		24-28 per Brtck	50	50
	NOTE: The bartack shall be centered on the belt loops.					
	e. The belt loops shall finish 1-1/2 (± 1/8) inches between tacks.					
18.	Bartack. Stitch a 3/8 to 7/16 inch long bartack as follows:	Brtck		21-28 per Brtck	50	50
	(1) One at base of fly across top ends of stitching of front crotch seam, superimposed on the end of "j" stitching (operation 16i.). (2) One superimposed on the "j" stitching of fly. The bottom end of the bartack shall be located 1 to 1 1/8 inches from the base of the fly, measured along the curve of the "j" stitching, relieving tension on the bottom stop. Bartack through all plies. (3) Two on each belt loop located as specified in operation 17. (4) One on the upper left and right corners of each cargo pocket, superimposed on the stitching. (5) One on the upper left and right corners of each cargo pocket flap, superimposed on the stitching at fold.					
19.	Cleaning. a. Remove all spots, stains and shade tickets.					
	b. Trim thread ends and remove all loose ends.					
	NOTE: Care shall be taken when trimming thread ends that damage to the fabric does not result.					

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TABLE I - CONSTRUCTION OF TROUSERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
20.	Pressing. a. Press legs smooth with creases along both inseams and outseams. b. Press top of trousers, including fly and ends of waistband.					

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3.8 Measurements. The finished trousers shall conform to the measurements shown in Table II.

Table II - Sizes and Measurements

Size	(A) Waist	(B) Seat	(C) Inseam Regular	(C) Inseam X-Long	(D) Bottom
26	26	34	35	39	19
27	27	35	35	39	19 1/4
28	28	36	35	39	19 1/2
29	29	37	35	39	19 3/4
30	30	38	35	39	20
31	31	39	35	39	20 1/4
32	32	40	35	39	20 1/2
33	33	41	35	39	20 3/4
34	34	42	35	39	21
35	35	43	35	39	21 1/4
36	36	44	35	39	21 1/2
37	37	45	35	39	21 3/4
38	38	46	35	39	22
40	40	48	35	39	22 1/4
42	42	50	35	39	23
44	44	52	35	39	23 1/2
46	46	54	35	39	24
48	48	56	35	39	24 1/2
50	50	58	35	39	25
Tolerance	± 1/2	± 1/2	± 1	± 1	± 1/2

NOTE: Refer to Figure 1 for A, B, C, D, and E measurements. Measurements shall be taken with the waistband snaps and the slide fastener fully closed. Trousers shall be laid out smooth and flat.

(A) Waist - Twice the measurement taken along the top edge of waistband from folded edge to folded edge.

(B) Seat - Twice the measurement taken 1 inch above the lower fly bartack from folded edge to folded edge.

(C) Inseam - Measurement shall be taken from crotch seam along the inseam to the unhemmed bottom edge.

(D) Bottom - Twice the measurement taken across the unhemmed bottom edge from folded edge to folded edge.

3.9 Workmanship. The finished trousers shall conform to the quality established by this specification. The occurence of defects shall not exceed the applicable acceptable point values or defect limits.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to acertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article submitted in accordance with 3.2 shall be inspected as specified in 4.4.2 for compliance with design, construction, workmanship, appearance and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-1488.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with the requirements for referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document. All test reports shall contain the individual values utilized in expressing the final result. The requirement that all components of the slide fastener have been furnished by one supplier shall be accepted on a certificate of compliance (see 3.3.3).

4.4.2 End item examination. Sampling and examination of the end item shall be performed in accordance with the provisions for utility trousers in MIL-STD-1488. In addition, the following defects also apply:

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	Point Value
Knee Patches	
1. Misplaced by more than 3/4 inch from pattern marks	1
2. Puckered, twisted, pleated or poorly shaped, affecting appearance	1
Snaps	
1. Any omitted	3
2. Any misplaced, not positioned or attached as specified	2
3. Finish omitted or not type specified	2
4. Any broken or maleformed	3
5. Any having a sharp or rough edge	2
6. Clinched	
a. Tightly, damaging or cutting adjacent material	3
b. Loosely, permitting any component to rotate freely, but not to the degree that any component can be expected to become detatched during use	2
c. Loosely, to the degree that any component can be expected to become detatched during use	3
Fastener Tape(s) (hook and loop)	
1. Omitted	3
2. Misplaced or not positioned as specified, i.e., not serving intended purpose	2
3. Type, color or width not as specified	1
4. Stitched too loose or too tight causing noticeable puckering or twist on outside	2

4.4.3 Packaging inspection. Packaging inspection shall be in accordance with MIL-STD-1488.

4.5 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5 as specified in MIL-STD-105.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

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5. PACKAGING

5.1 Preservation-packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each pair of pressed trousers (see Table I, operation 20) shall be folded with fly inwards, laying one leg atop the other, matching side seams. Fold trousers in half by bringing up bottom edge even with top of waistband, and in half again by bringing the folded knee edge up even with top edge of waistband. The folded trouser shall measure approximately 12 by 14 inches.

5.1.2 Level C. Trousers shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Fourteen (14) pairs of trousers of one size and length only, packaged as specified in 5.1, shall be stacked in two piles side by side, with waistband alternated side for side, and packed in a fiberboard shipping container, assembled, closed and reinforced conforming to type CF, class weather-resistant, variety DW, grade V15c size 2A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic of MIL-B-17757. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper.

5.2.2 Level B. Fourteen (14) pairs of trousers of one size and length only, packaged as specified in 5.1, shall be stacked in two piles side by side with waistband alternated side for side, and packed in a fiberboard shipping container assembled, closed and reinforced conforming to type CF, class domestic, variety DW, grade 200, size 2A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic of MIL-B-17757. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper.

5.2.3 Level C. Item, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the US Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.2.4 Palletization. When specified (see 6.2) item, packed as specified shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the bonding means C, K, and L or O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147.

The pallet shall be 4-way, type IV; type V, class 1, size 2; or type VIII, fabricated from wood groups I, II, III, or IV, grade A of NN-P-71, or 4-way, style 1, size A, type I, class 1 fabricated from wood groups specified, of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern

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of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.3 Marking. In addition to any special marking required by the contract or order, unit packs, intermediate or exterior containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

5.4 MIL-STD-2073. When specified (see 6.2) Preservation and Packing shall be in accordance with MIL-STD-2073-1 and MIL-STD-2073-2.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The trousers are intended to be worn in conjunction with the flight deck jersey in the performance of shipboard flight deck duties.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Sizes and lengths required (see 1.2).
c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article inspection is required (see 3.2), the item will be tested and shall be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangement for examination, quantity, and testing and approval.
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
f. When palletization is required (see 5.2.4).
g. When all packaging data is to be found in MIL-STD-2073-1 and MIL-STD-2073-2.

6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids (see 3.1 and 3.5).

6.4 Subject term (key word) listing.
Clothing, flight deck
Denim
Flame retardant treated
Knee patch
Trouser, utility
Uniform, utility

Custodian:
Navy - NU

Preparing activity:
Navy - NU

Review activity:
DLA - CT

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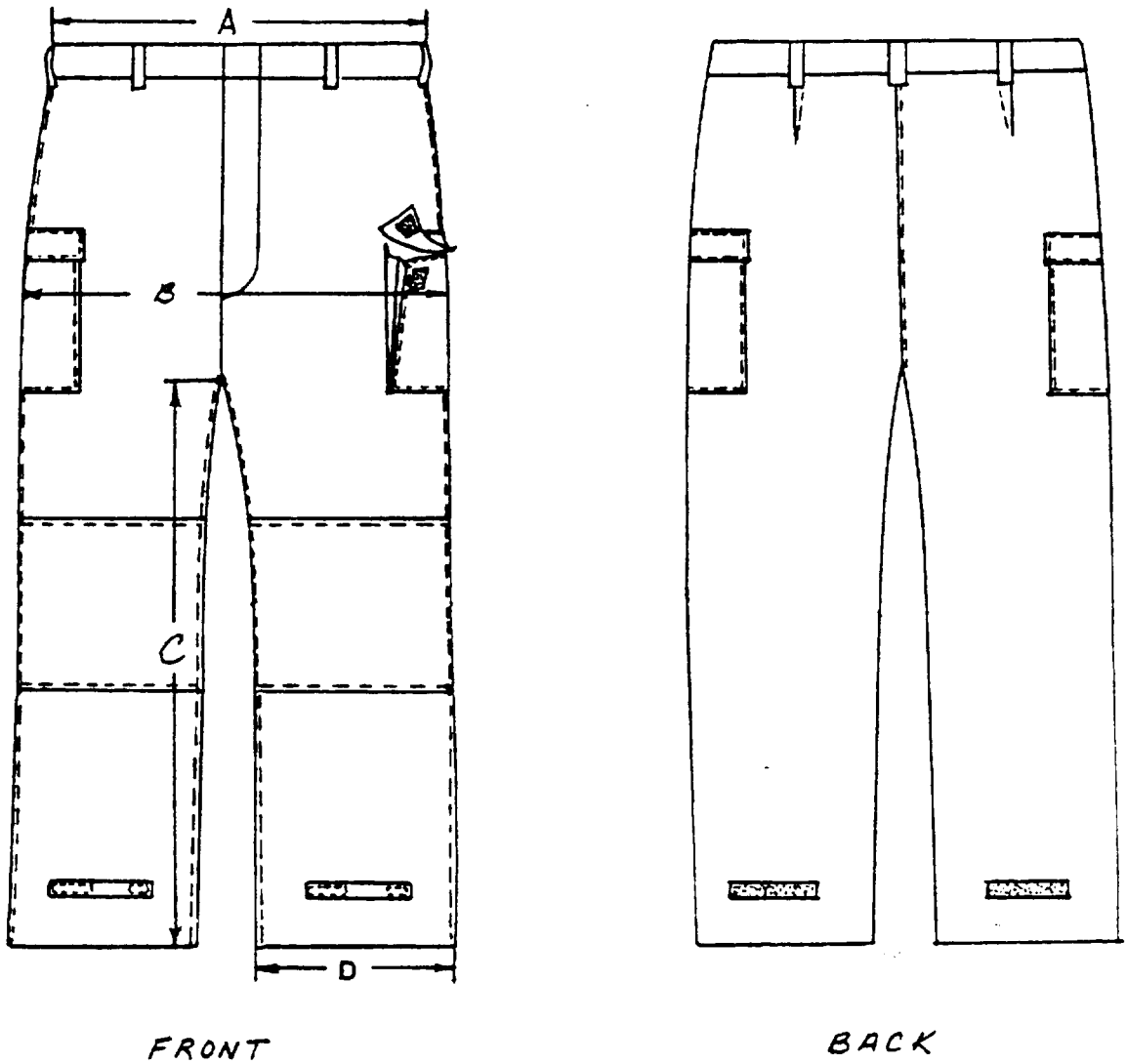


Figure 1 - Trousers, Flight Deck, Crewmen's

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-24944(NU)	TROUSERS, FLIGHT DEC & CREWMEN'S

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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